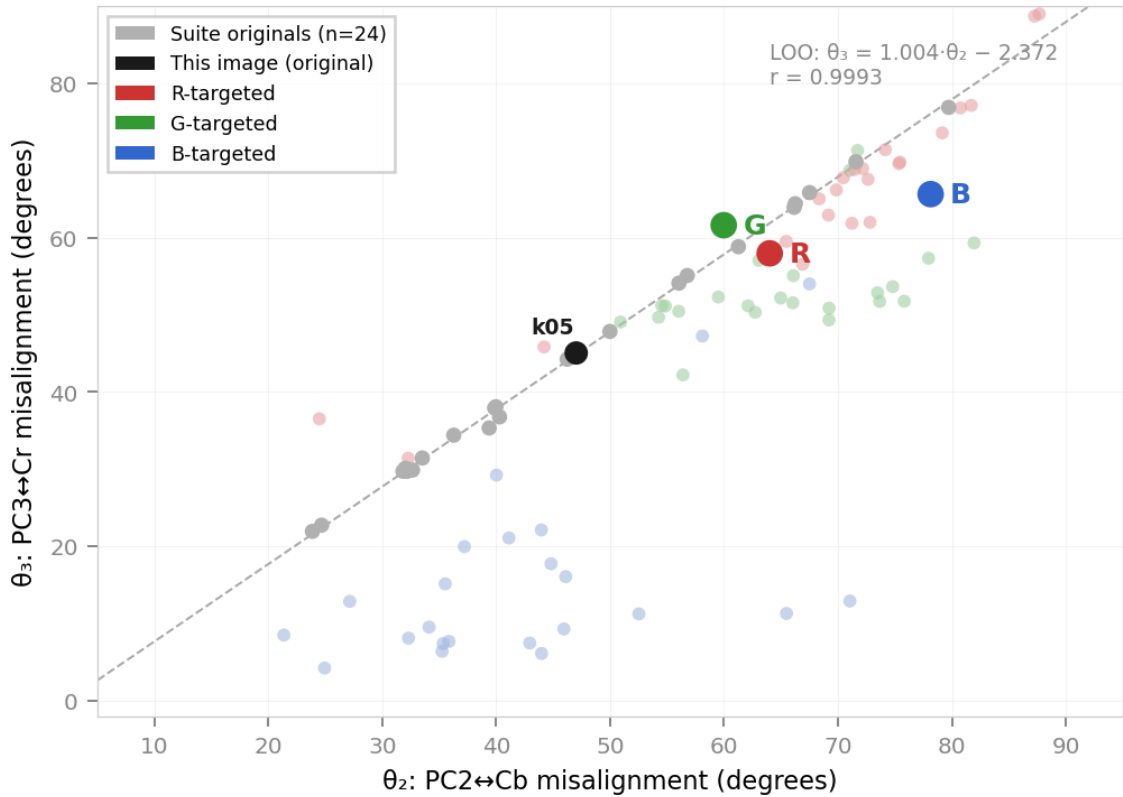


Compression Profile: KODIM05768x512 | 24-bit RGB | PNG (lossless)

motocross bikes | Weakly One-Dimensional (PC1 = 91.91%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.8981	0.8215	0.9034	0.874	47.02°	45.07°	+0.23°
R-targeted	0.1926	0.0660	0.9701	0.409	64.01°	57.99°	-3.90°
G-targeted	-0.0583	0.9240	-0.0778	0.353	59.97°	61.64°	+3.80°
B-targeted	0.9325	-0.0401	0.0251	0.333	78.14°	65.67°	-10.41°

R-targeted isolates red (R-G, R-B < 0.19) while preserving G-B coupling at 0.970.
G-targeted isolates green (R-G, G-B < 0.08) while preserving R-B at 0.924.
B-targeted preserves R-G at 0.932 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM05768x512 | 24-bit RGB | PNG (lossless)

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3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

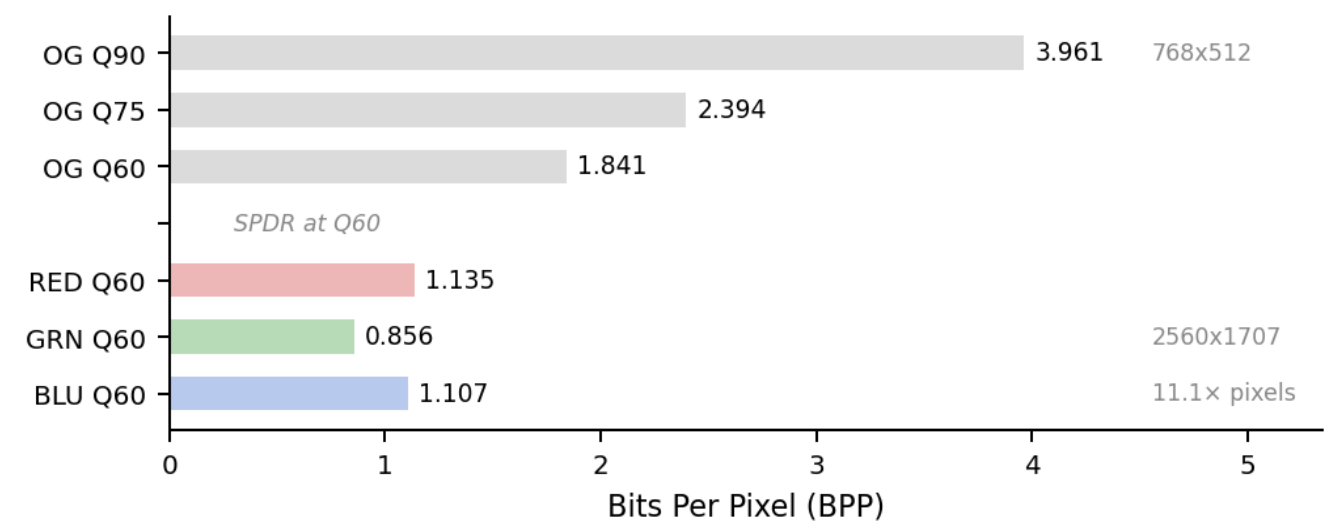
ORIGINAL (kodim05)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	767.2 KB	15.983	393,216
JPEG Q90	JPEG	768x512	190.1 KB	3.961	393,216
JPEG Q75	JPEG	768x512	114.9 KB	2.394	393,216
JPEG Q60	JPEG	768x512	88.4 KB	1.841	393,216

REDISTRIBUTED (kodim05 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	605.3 KB	1.135	4,369,920
GRN Q60	JPEG	2560x1707	456.5 KB	0.856	4,369,920
BLU Q60	JPEG	2560x1707	590.5 KB	1.107	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM05768x512 | 24-bit
RGB | PNG (lossless)

motocross bikes | Weakly One-Dimensional (PC1 = 91.91%)

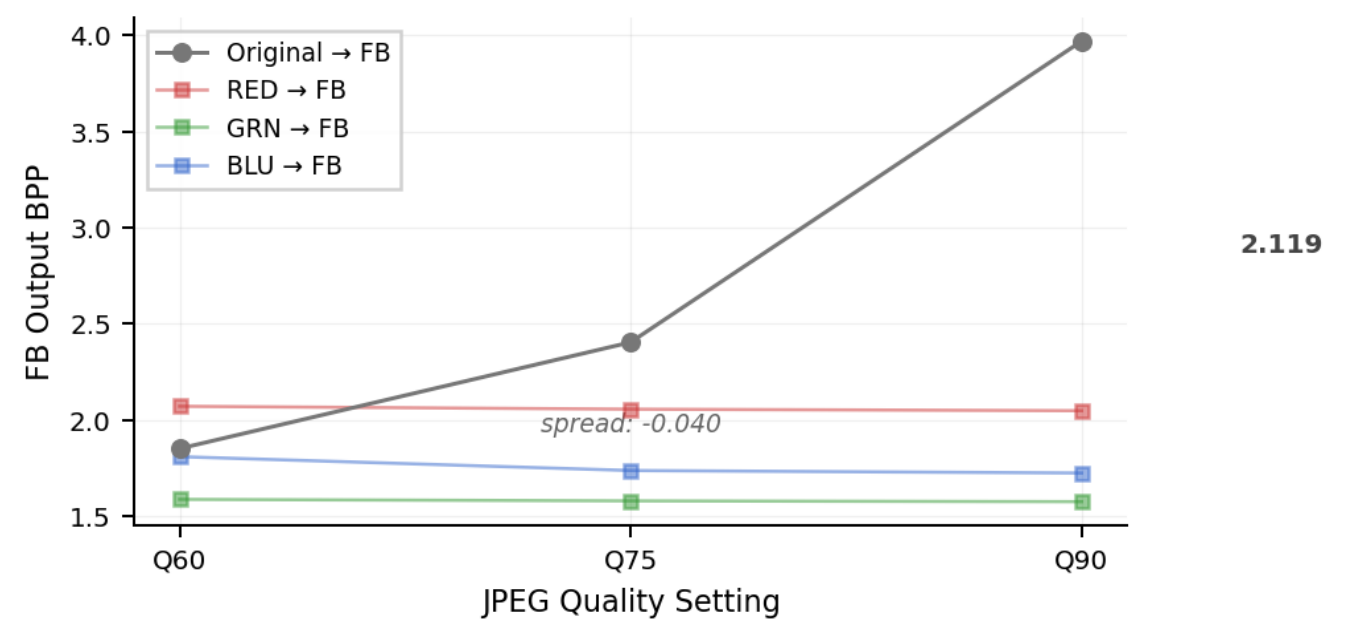
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	3.337	–
R-targeted → FB	2560x1707	2048x1366	2.048	38.6%
G-targeted → FB	2560x1707	2048x1366	1.575	52.8%
B-targeted → FB	2560x1707	2048x1366	1.724	48.3%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.349	59.6%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.8729 vs 0.874 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 2.119 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.040 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim05

Best BPP (Q60, direct)	0.856	2560x1707, 4,369,920 px
Best BPP through FB	1.575	2048x1366 output
vs OG Q90 → FB	60.3%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216